

MDM-ASR: Bridging Accuracy and Efficiency in ASR with Diffusion-Based Non-Autoregressive Decoding

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**Università
degli Studi
di Palermo**

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

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Experimental Setup

Experimental Results

Conclusion

Motivation

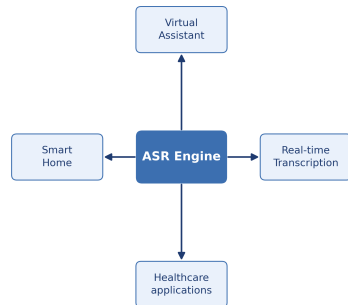
ASR is a key enabling technology for numerous applications.

The trade-off

Transformer-based seq2seq ASR with cross-attention faces a fundamental accuracy–efficiency trade-off:

	AR	NAR
Accuracy	High	Lower
Decoding speed	Slow (sequential)	Fast (parallel)

Goal: combine AR-level accuracy with NAR-level efficiency.



Examples of ASR applications

Proposed Masked Diffusion Model ASR

Goal: Bridge the AR–NAR performance gap while keeping parallel decoding efficiency.

Model

- ▶ **Pre-trained speech encoder** for acoustic representations
- ▶ **Discrete diffusion decoder** refining masked tokens via cross-attention

Training & Inference

- ▶ **Diffusion loss** from the forward corruption process
- ▶ **Iterative Self-Correction Training:** reduces train–inference mismatch
- ▶ **Position-biased EB-Conf sampler:** stable, efficient decoding

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

Diffusion Models

Key idea

Diffusion models learn to reverse a forward corruption process $q(\mathbf{z}_t|\mathbf{x})$ that gradually transforms clean data $\mathbf{x} \sim q(\mathbf{x})$ into noisy latent variables $\mathbf{z}_t$, $t \in [0, 1]$.

Continuous diffusion

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{x} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ & α_t is a monotonically decreasing noise schedule.

Discrete diffusion

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi}),$$

where $\boldsymbol{\pi}$ is the target vector & α_t is a monotonically decreasing noise schedule.

Masked Diffusion Process

Absorbing state

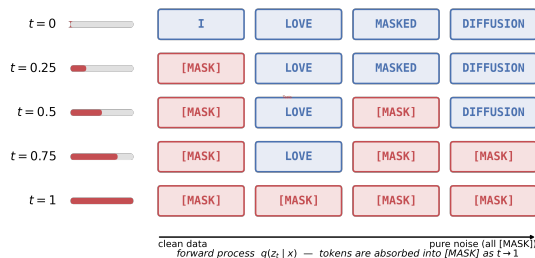
For $\pi = \mathbf{m}$: masked diffusion process, with transition probability

$$q(\mathbf{z}_t | \mathbf{z}_s) = \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \mathbf{m})$$

where $\alpha_{t|s} = \alpha_t / \alpha_s$, $0 < s < t < 1$

- ▶ Once masked, always masked: absorbing state
- ▶ At $t = T$: all inputs masked (prob. 1)

Forward Diffusion Process



Reverse Unmasking Process

Posterior Denoising Distribution

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t), & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1-\alpha_t}\right), & \mathbf{z}_t = \mathbf{m}. \end{cases}$$

Neural Approximation

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t)),$$

where $\mathbf{x}$ is replaced by the model prediction $\mathbf{x}_\theta(\mathbf{z}_t, t)$.

- ▶ Unmasked tokens remain unchanged.
- ▶ Only masked tokens are refined.
- ▶ Clean input tokens are never re-masked.

Sequence-Level Training Objective

- ▶ Corrupt each token **independently**: $\mathbf{x}^{(1:L)} \rightarrow \mathbf{z}_t^{(1:L)}$
- ▶ Train to **maximize likelihood** of clean tokens under the reverse model

Objective

$$\mathcal{L} = \mathbb{E}_q \int_0^1 \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell=1}^L \log \left\langle \mathbf{x}_{\theta}^{(\ell)}(\mathbf{z}_t^{(1:L)}, t), \mathbf{x}^{(\ell)} \right\rangle dt$$

$\mathbf{x}_{\theta}^{(\ell)}(\cdot)$: predicted probability at position ℓ $\alpha'_t = d\alpha_t/dt$

Key property

Reconstructs masked tokens at **all noise levels** $\Rightarrow$ fully **non-autoregressive** generation

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

MDM-ASR: Audio Conditioning

Audio-conditioned Masked Diffusion Model

- ▶ Condition on the discrete denoising process on the acoustic input, $\mathbf{a} \in \mathbb{R}^{T \times D}$, from a pretrained speech encoder.
- ▶ Forward corruption process independent of the audio signal.
- ▶ The reverse unmasking process is conditioned on the acoustic representation:

$$p_{\theta}(\mathbf{z}_s^{(1:L)} | \mathbf{z}_t^{(1:L)}, \mathbf{a}) = \prod_{l=1}^L p_{\theta}(\mathbf{z}_s^{(l)} | \mathbf{z}_t^{(1:L)}, \mathbf{a})$$

MDM-ASR: The Reverse Process

Denosing with acoustic conditioning

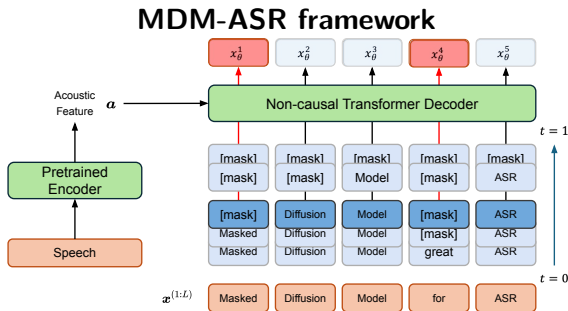
- ▶ Let $\mathbf{x}_{\theta}^{(l)}(\mathbf{z}_t^{(1:L)}, a) \in \Delta^K$ be the model-predicted categorical distribution at position l .
- ▶ The denoising posterior distribution conditioned on the audio is given by:

$$p_{\theta}(\mathbf{z}_s^{(1:L)} | \mathbf{z}_t^{(1:L)}, a) = \begin{cases} \text{Cat}(\mathbf{z}_s^{(l)}; \mathbf{z}_t^{(l)}), & \mathbf{z}_t^{(l)} \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s^{(l)}; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}_{\theta}^{(l)}(\mathbf{z}_t^{(1:L)}, \mathbf{a})}{1-\alpha_t}\right), & \mathbf{z}_t^{(l)} = \mathbf{m}. \end{cases}$$

MDM-ASR: Model Architecture

Key Components

- ▶ Encoder-decoder ASR with a **diffusion-based decoder**
- ▶ **Pre-trained speech encoder** for acoustic representations
- ▶ **Cross-attention** conditioning on acoustic features
- ▶ **Non-causal self-attention** over the full sequence
- ▶ **Parallel refinement** of all output tokens per diffusion step



MDM-ASR: Training Objective

- ▶ The noise schedule is defined as:

$$\alpha_t = 1 - t$$

- ▶ The training objective is:

$$\mathcal{L} = \mathbb{E}_q \int_0^1 -\frac{1}{t} \sum_{\ell: \mathbf{z}_t^{(\ell)} = \mathbf{m}} \log \left\langle \mathbf{x}_\theta^{(\ell)}(\mathbf{z}_t^{(1:L)}, \mathbf{a}), \mathbf{x}^{(\ell)} \right\rangle dt.$$

MDM-ASR: Iterative Self-Correction Training

Motivation

MDMs suffer from a train–inference mismatch:

- ▶ **Training:** denoise oracle corrupted sequences.
- ▶ **Inference:** refine self-generated sequences.

ISCT

Exposes the model to its own predictions.

1. Corrupt transcript: $\mathbf{z}_{t_1} \sim q(\mathbf{z}_{t_1} | \mathbf{x})$
2. First prediction: $\hat{\mathbf{x}}^{(1)} = \mathbf{x}_\theta(\mathbf{z}_{t_1}, \mathbf{a})$
3. Self-corrupt prediction: $\hat{\mathbf{z}}_{t_2} \sim q(\hat{\mathbf{z}}_{t_2} | \hat{\mathbf{x}}^{(1)})$
4. Refine prediction: $\hat{\mathbf{x}}^{(2)} = \mathbf{x}_\theta(\hat{\mathbf{z}}_{t_2}, \mathbf{a})$

$$\mathcal{L}_{\text{ISCT}} = \mathbb{E}_q \int_0^1 -\frac{1}{t_1} \sum_{\ell | \mathbf{z}_{t_1}^\ell = m} \log \langle \mathbf{x}_\theta^{(\ell)}(\mathbf{z}_{t_1}^{(1:L)}, \mathbf{a}), \mathbf{x}^{(\ell)} \rangle dt_1 + \mathbb{E}_q \int_0^1 -\frac{1}{t_2} \sum_{\ell | \hat{\mathbf{z}}_{t_2}^\ell = m} \log \langle \mathbf{x}_\theta^{(\ell)}(\hat{\mathbf{z}}_{t_2}^{(1:L)}, \mathbf{a}), \mathbf{x}^{(\ell)} \rangle dt_2.$$

PBEB-Conf: Position-Biased EB-Conf Sampler

- ▶ **Motivation:** Incorporate a **positional trajectory prior** into EB-Conf
 - ▶ Earlier tokens provide context that disambiguates later ones
 - ▶ $\Rightarrow$ decoding **left-to-right first** is often preferable
- ▶ **Limitation of EB-Conf:** Position-agnostic
 - ▶ Two equally confident positions treated the same, regardless of location
- ▶ **Positional bias:** $P_i = e^{-\lambda i}$, i = token index
 - ▶ Larger $i \rightarrow$ smaller P_i (later positions penalized)
 - ▶ λ controls bias strength; $\lambda = 0 \Rightarrow P_i = 1$ (no bias)
- ▶ **Reweighted confidence:** $\hat{c}_i = P_i \cdot c_i$
 - ▶ c_i = model's confidence at position i
 - ▶ Favors earlier positions in the ranking
- ▶ **Sampling rule:** Entropy-bounded criterion applied to positions **re-ranked** by $\hat{c}_i$

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

Datasets: English Benchmarks

- ▶ **LibriSpeech**: read audiobook speech, test-clean/test-other splits
- ▶ **Earnings22**: earnings calls, diverse speakers/accents, financial discourse
- ▶ **AMI**: multi-party meetings, word-aligned manual transcriptions
- ▶ **VoxPopuli**: European Parliament recordings, 20+ languages

Dataset	Domain	Train (h)	Val (h)	Test (h)
LibriSpeech	Audiobook	960	11	10.74
Earnings22	Financial meetings	105	5	5.43
AMI	Meetings	78	5	8.54
VoxPopuli	European parliament	523	5	4.93

Datasets: Multilingual (MLS)

- ▶ **Multilingual LibriSpeech (MLS)** benchmark
- ▶ Three languages covered by the pretrained model: **German, Spanish, French**
- ▶ Total training data: **~4,000 hours**

Language	Train (h)	Val (h)	Test (h)
German	1966.51	14.28	14.29
Spanish	917.68	9.99	10
French	1076.58	10.07	10.07

Evaluation Metrics

Accuracy: Word Error Rate (WER)

- ▶ Transcripts and predictions normalized with the **Whisper Normalizer**

Efficiency: Inverse Real-Time Factor (RTF_x)

$$\text{RTF}_x = \frac{\text{Total audio duration (s)}}{\text{Decoding time (s)}}$$

- ▶ $\text{RTF}_x > 1$: faster than real time; higher is better
- ▶ Single **NVIDIA A100** GPU, batch size 1, full precision, no compilation

Implementation Details

- ▶ Built on **NVIDIA NeMo**, initialized from pretrained canary-1b-flash ($\sim 1\text{B}$ params)
- ▶ Decoder self-attention: **causal** $\rightarrow$ **non-causal** for masked diffusion decoding
- ▶ **Adam** optimizer, learning rate 5×10^{-4} , batch size 8
- ▶ Two-step **ISCT** schedule (extends naturally to more steps)
- ▶ Training: pad short sequences with EOS, treated as a standard token
- ▶ Inference: truncate output at first EOS $\rightarrow$ improves stability & efficiency
- ▶ Max decoding length: 256 tokens

Baselines

Compared against top models from the **HF Open ASR Leaderboard**, spanning AR, NAR, and diffusion-based ASR.

Category	Models
AR / Encoder-decoder LLM-augmented	Whisper, Canary-1b-flash, SeamlessM4T-large-v2 Phi-4-multimodal, Qwen2-Audio, Voxtral
NAR (CTC-based)	OWSM-CTC, Parakeet-CTC, XLSR-53
NAR (Diffusion/Flow)	Transfusion, Whisfusion, FFDM, Drax

- Official evaluation scripts used to **reproduce all baselines** for a fair, consistent comparison

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

Results: English Benchmarks

	Model	LS Clean	LS Other	WER (%) $\downarrow$			Avg.	Params (B)	RTFx $\uparrow$
				Earnings22	AMI	VoxPopuli			
AR	Whisper-large-v3	2.0	3.9	11.3	16.0	9.5	8.5	1.5	12.83
	OWSM-v3.1 $\dagger$	2.4	5.0	15.4	20.4	8.4	10.3	1.0	15.52
	Canary-1b-flash	1.5	2.9	12.8	13.1	5.6	7.2	1.0	29.25
	Phi-4-multimodal	1.7	3.8	10.2	11.7	6.0	6.7	5.6	4.85
	Qwen2-Audio $\dagger$	1.6	3.6	14.1	15.2	7.1	8.3	8.4	4.18
	Voxtral-Mini	1.9	4.1	10.7	16.3	7.1	8.0	3.0	12.19
NAR	OWSM-CTC $\dagger$	2.4	5.2	17.2	23.8	8.6	11.4	1.0	132.54
	Parakeet-CTC	1.8	3.5	13.8	15.7	6.6	8.3	1.1	120.20
	TransFusion $\dagger$	6.7	8.8	-	-	-	-	0.2	0.57
	Whisfusion $\dagger$	8.3	17.0	-	-	-	-	0.3	84.86
	FDDM $\dagger$	4.0	7.2	-	-	-	-	0.6	-
	Drax $\dagger$	2.6	5.7	15.2	13.9	8.6	9.2	1.2	11.32
	MDM-ASR (ours)	1.8	3.6	10.7	12.2	6.0	6.9	1.0	46.81

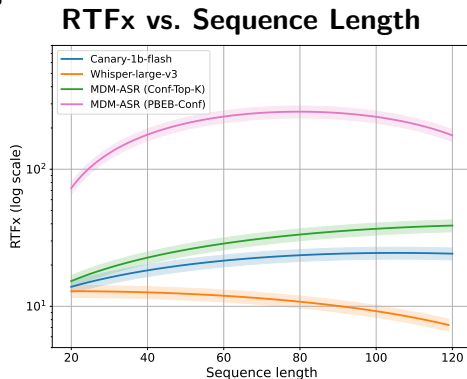
Results: Multilingual (MLS)

Model	DE	ES	FR
Whisper-large-v3	-	3.0	4.8
Phi-4-multimodal	4.8	3.6	4.9
Voxtral-Mini	7.1	5.1	5.3
Canary-1b-flash	4.4	-	4.5
OWSM-CTC	11.9	10.3	12.9
XLSR-53	6.5	6.1	5.6
Drax	7.7	5.4	7.1
MDM-ASR (ours)	3.6	2.9	3.8

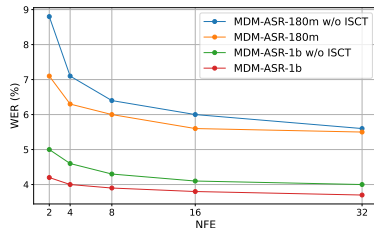
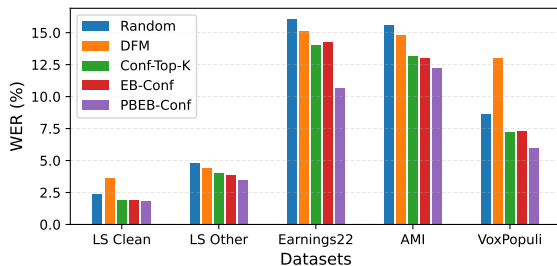
- ▶ **New SOTA** among generative NAR models: beats Drax, OWSM-CTC, XLSR-53 on all 3 languages
 - ▶ On-par or better than **AR** models: beats Whisper-large-v3 on ES/FR
 - ▶ Beats **LLM-augmented** models: Phi-4-multimodal, Voxtral-Mini
 - ▶ vs. **Canary-1b-flash** (same encoder/training): gains on DE (3.6 vs 4.4) and FR (3.8 vs 4.5)
- Gains reflect **masked diffusion decoding**, not just language coverage

Decoding Efficiency

- ▶ **AR** (e.g. Whisper-large-v3): RTF_x declines steadily with sequence length (sequential decoding)
- ▶ **Canary-1b-flash**: roughly constant, limited scaling benefit
- ▶ **MDM-ASR**: consistently higher RTF_x, gap **widens** for longer outputs
- ▶ **EB-Conf / PBEB-Conf**: highest RTF_x overall, large margin over AR
- ▶ PBEB-Conf slightly declines at long sequences (adaptive update → converges to Conf-Top- K), still faster than AR



Ablation Studies



ISCT effect (LS Other):

- Improves WER across **all** NFE budgets, both model sizes
- Largest gains at **low NFE** (2, 4, 8 steps)
- Smaller but consistent gains for 1B model
- **No extra inference cost**

Sampler choice:

- Random-Unmasking, DFM: Suboptimal
- Confidence-based samplers: Strong & Consistent
- EB-Conf: Same accuracy, faster decoding
- PBEB-Conf: Additional gains from positional bias

Outline

Introduction

Background

Proposed Masked Diffusion Based ASR

Experimental Setup

Experimental Results

Conclusion

Conclusion

- ▶ **MDM-ASR**: audio-conditioned masked diffusion framework for NAR ASR
 - ▶ Standard encoder–decoder architecture
 - ▶ Sequential decoding → iterative parallel unmasking
- ▶ **ISCT** and **position-biased EB sampler**: improve training & inference efficiency
- ▶ **Results** (English & multilingual benchmarks):
 - ▶ Outperforms prior generative **NAR** models
 - ▶ Narrows the gap to strong **AR** models
 - ▶ Favorable decoding efficiency
- ▶ Masked diffusion: a **competitive and practical** direction for NAR ASR

Thank you!

Questions?